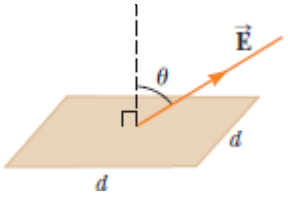
© 2026 Cengage Learning, Inc., Cengage Learning, Inc.

Chapter Review

Problems: Additional Problems

37. Find the electric flux through the plane surface shown in Figure P23.37 if $\theta = 60.0°$, $E = 350\ \mathrm{N/C}$, and $d = 5.00\ \mathrm{cm}$. The electric field is uniform over the entire area of the surface.

Figure P23.37



38. Three solid plastic cylinders all have radius 2.50 cm and length 6.00 cm. Find the charge of each cylinder given the following additional information about each one. Cylinder

(a) carries charge with uniform density $15.0\ \mathrm{nC/m^2}$ everywhere on its surface. Cylinder

(b) carries charge with uniform density $15.0\ \mathrm{nC/m^2}$ on its curved lateral surface only. Cylinder

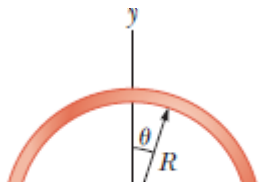
(c) carries charge with uniform density $500\ \mathrm{nC/m^3}$ throughout the plastic.

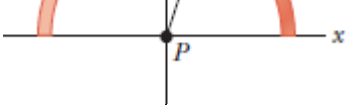
39. S A line of charge starts at $x = +x_0$ and extends to positive infinity. The linear charge density is $\lambda = \lambda_0 x_0 / x$, where λ_0 is a constant. Determine the electric field at the origin.

40. S Show that the maximum magnitude $E_{\max}$ of the electric field along the axis of a uniformly charged ring of radius a and total charge Q occurs at $x = a\sqrt{2}$ (refer to Fig. 23.3) and has the value $Q/(6\sqrt{3}\pi\varepsilon_0 a^2)$.

41. A line of positive charge is formed into a semicircle of radius $R = 60.0\ \mathrm{cm}$ as shown in Figure P23.41. The charge per unit length along the semicircle is given by the expression $\lambda = \lambda_0 \cos\theta$. The total charge on the semicircle is $12.0\ \mu\mathrm{C}$. Calculate the total force on a charge of $3.00\ \mu\mathrm{C}$ placed at the center of curvature P.

Figure P23.41





42. S A very large conducting plate lying in the xy plane carries a charge per unit area of σ. A second such plate located above the first plate at $z = z_0$ and oriented parallel to the xy plane carries a charge per unit area of -2σ. Find the electric field for

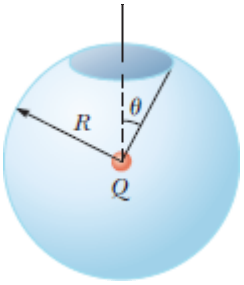
(a) $z < 0$,

(b) $0 < z < z_0$, and

(c) $z > z_0$.

43. NS A sphere of radius $R = 1.00\ \text{m}$ surrounds a particle with charge $Q = 50.0\ \mu\text{C}$ located at its center as shown in Figure P23.43. Find the electric flux through a circular cap of half-angle $\theta = 45.0°$.

Figure P23.43



44. NS A sphere of radius R surrounds a particle with charge Q located at its center as shown in Figure P23.43. Find the electric flux through a circular cap of half-angle θ.

Chapter 23: Continuous Charge Distributions and Gauss's Law Problems: Additional Problems
Book Title: Physics for Scientists and Engineers with Modern Physics
Printed By: Pragun Damani (pd2752@nyu.edu)
© 2026 Cengage Learning, Inc., Cengage Learning, Inc.

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may by reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.